Structured notes: Module 4, transient circuits and alternating currents

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

Section 1: growth and decay of current in an L-R circuit (Murugeshan 12.1, 12.2)

12.1 Growth of current in an L-R circuit

Source: book-sources/m4-s12-1-growth.html, Murugeshan pp. 312–313 (printed). Syllabus item 15. Equations (1)–(4) are given in full.

Prerequisites

- Ohm's law and the emf of a source (Plus Two; pre-physics P1).
- First-order differential equations solved by separation of variables (Beyond; pre-math M4).
- Exponentials and natural logarithms (pre-math M3).

1. Setup

An inductance L and a resistance R are connected in series with a steady emf E through a key (Fig. 12.1). On closing the key the current rises from zero to a final value I_0. I denotes the current at time t.

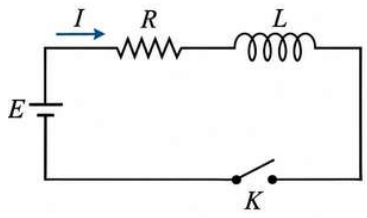


Fig. 12.1 (book): inductance L and resistance R in series with a cell E and key K. Schematic.

2. Loop equation

The induced back emf is $\varepsilon = -L\,dI/dt$. The loop equation is

$$E = RI + L\frac{dI}{dt} \qquad (12.1(1))$$

3. Final current

At the maximum, $dI/dt = 0$, so

$$E = RI_0 \qquad (12.1(2))$$

> **RECALL (PLUS TWO)** Ohm's law gives $I_0 = E/R$ for a steady current, which is the value the inductor allows once its back emf vanishes.

4. Variables separated

Substituting Eq. (2) into Eq. (1) gives $R(I_0 - I) = L\,dI/dt$, that is

$$\frac{dI}{I_0 - I} = \frac{R}{L}\,dt$$

5. Integration and boundary condition

Integration gives

$$-\log(I_0 - I) = \frac{R}{L}t + C \qquad (12.1(3))$$

The boundary condition $I = 0$ at $t = 0$ fixes $C = -\log I_0$, so $\log_e\left[(I_0 - I)/I_0\right] = -(R/L)t$.

6. Solution

$$I = I_0\left(1 - e^{-(R/L)t}\right) \qquad (12.1(4))$$

7. Time constant

The time constant is $\tau = L/R$. At $t = \tau$, $I = (1 - e^{-1})I_0 = 0.632\,I_0$. The current reaches $0.865\,I_0$ at 2τ and $0.950\,I_0$ at 3τ (Fig. 12.2).

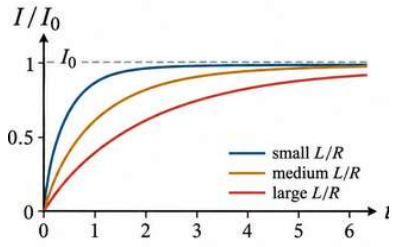


Fig. 12.2 (book): current I/I0 against time for low, medium and large L/R. Drawn from I = I0(1 − e^(−Rt/L)).

8. Worked example (printed p. 314)

For $R = 10\ \Omega$, $L = 0.5$ H and $E = 10$ V, $R/L = 20\ \text{s}^{-1}$. The time to reach 63.2% of I_0 is $t = \frac{\ln 2.717}{20} = 0.05$ s, which equals $\tau = L/R$.

12.2 Decay of current in an L-R circuit

Source: book-sources/m4-s12-2-decay.html, Murugeshan pp. 313–314 (printed), with Examples 1 and 2. Syllabus item 16. Equations (1) and (2) are given in full. Example 1 is worked in §12.1.

Prerequisites

- §12.1 of these notes: the loop equation and the time constant.
- Exponentials and natural logarithms (pre-math M3).

1. Setup

The cell is disconnected and the two ends of the circuit are joined. The emf is zero, and the inductor drives the current through R by its back emf.

2. Loop equation

$$0 = RI + L\frac{dI}{dt} \qquad (12.2(1))$$

Separating variables, $dI/I = -(R/L)\,dt$.

3. Solution

Integration gives $\log_e I = -(R/L)t + C$. The initial condition $I = I_0$ at $t = 0$ gives $C = \log_e I_0$, and so

$$I = I_0 e^{-(R/L)t} \qquad (12.2(2))$$

4. Interpretation

The decay time constant is L/R, the time in which the current falls to $1/e$ of I_0. The rate of decay is $dI/dt = -(R/L)I$. Growth and decay are complementary (Fig. 12.5).

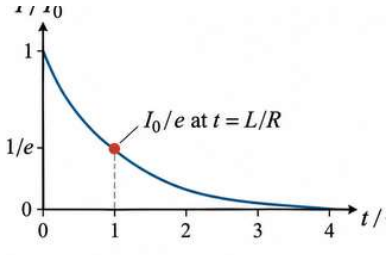


Fig. 12.3 (book): decay of current I = I0 e^(−Rt/L) against time. At t = L/R the current is I0/e.

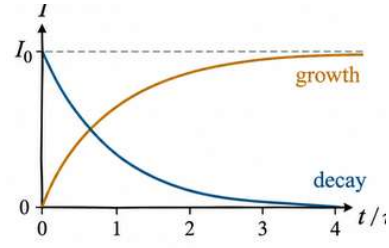


Fig. 12.5 (book): growth and decay curves shown together; they are complementary.

5. Worked example (printed p. 315)

$I_0 = E/R = 2$ A and $R/L = 10\ \text{s}^{-1}$, so $I = 2e^{-10t}$. At $t = 0.05$ s, $I = 1.213$ A. At $t = 0.2$ s, $I = 0.271$ A.

Section 2: charge in C-R and L-C circuits; leakage (Murugeshan 12.3 to 12.5)

12.3 Growth and decay of charge in a C-R circuit

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 315–317 (printed). Syllabus items 15 and 16. Equations (a)(1)–(a)(3) and (b)(1)–(b)(3) are given in full.

Prerequisites

- $Q = CV$ and the p.d. across a capacitor (pre-physics P3).
- First-order equations solved by separation of variables (pre-math M4).

1. Charging

> **RECALL (PLUS TWO)** $Q = CV$, so the p.d. across the capacitor is Q/C. The emf is shared between the capacitor and the resistor in series.

A capacitor C and a resistance R are connected to an emf E through a key (Fig. 12.6). Q denotes the charge on C at time t, and $I = dQ/dt$.

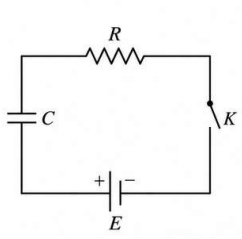


Fig. 12.6 (book): capacitor C, resistance R, Morse key K and cell E in series.

$$E = \frac{Q}{C} + RI \qquad \text{(12.3(a)(1))}$$

2. Final charge

At the maximum charge Q_0, $dQ/dt = 0$ and $E = Q_0/C$.

3. Variables separated

$$(Q_0 - Q) = CR\frac{dQ}{dt}$$
$$\frac{dQ}{Q_0 - Q} = \frac{dt}{CR} \qquad \text{(12.3(a)(2))}$$

4. Solution

With $Q = 0$ at $t = 0$,

$$Q = Q_0\left(1 - e^{-t/CR}\right) \qquad \text{(12.3(a)(3))}$$

The time constant is CR. At $t = CR$ the charge is $0.632\,Q_0$.

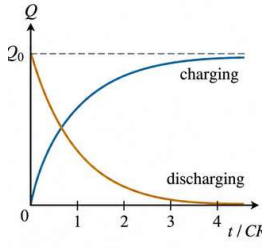


Fig. 12.7 (book): charge against time; growth to Q_0 and discharge from Q_0 are complementary.

5. Discharging

With $E = 0$ and the capacitor charged to Q_0,

$$R\frac{dQ}{dt} + \frac{Q}{C} = 0 \qquad \text{(12.3(b)(1))}$$

$$Q = Q_0 e^{-t/CR} \qquad \text{(12.3(b)(2))}$$

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR}e^{-t/CR} = -\frac{Q}{CR} \qquad \text{(12.3(b)(3))}$$

At $t = CR$, $Q = 0.368\,Q_0$; the time constant is also the time to fall to $1/e$.

6. Worked examples (printed pp. 316–317)

(1) With $R = 2\ \text{M}\Omega$ and $t = 0.5$ s for $Q/Q_0 = 3/4$, Eq. (a)(3) gives $C = 0.18\ \mu$F. (2) For $C = 0.1\ \mu$F and $R = 10\ \text{M}\Omega$, the p.d. halves in $CR\ln 2 = 0.6931$ s.

12.4 Measurement of high resistance by leakage (background)

Background. *Outside the Module 4 items (15–22). Kept because it applies the discharge law of §12.3. The book numbers no equations in this section. Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 318–319 (printed).*

1. Principle

A capacitor of capacitance C and initial charge Q_0 discharges through a resistance R for a time t. From §12.3 the remaining charge is $Q = Q_0 e^{-t/CR}$, so

$$R = \frac{t}{C \log_e (Q_0/Q)} = \frac{t}{2.3026\, C \log_{10} (Q_0/Q)}$$

A high R gives a slow discharge. Measuring Q_0/Q then determines R.

2. Circuit and procedure

C is a capacitor of known value, R the high resistance, B.G. a ballistic galvanometer, E a cell, and K_1, K_2, K_3 tap keys (Fig. 12.8).

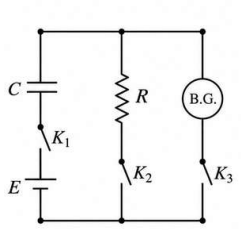


Fig. 12.8 (book): capacitor C, high resistance R, ballistic galvanometer B.G., cell E and keys K1, K2, K3.

(i) The capacitor is charged through K_1 and then discharged through the galvanometer, giving a throw $\theta_0 \propto Q_0$. (ii) The capacitor is charged again, allowed to leak through R for a time t, and then discharged through the galvanometer, giving a throw $\theta \propto Q$. Since $Q_0/Q = \theta_0/\theta$,

$$R = \frac{t}{2.3026\, C \log_{10} (\theta_0/\theta)}$$

3. Graph

$\log_{10}(\theta_0/\theta)$ plotted against t is a straight line through the origin (Fig. 12.9). Its slope gives $t/\log_{10}(\theta_0/\theta)$, which with C known gives R.

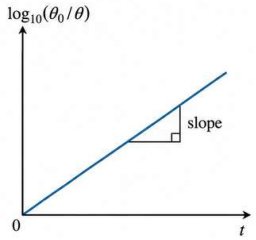


Fig. 12.9 (book): straight line of log10(θ0/θ) against t; the slope gives t / log10(θ0/θ).

4. Worked examples (printed pp. 318–319)

(1) $C = 2\ \mu$F, $R = 100\ \text{M}\Omega$, $Q_0/Q = 2$: $t = 2.3026\, CR \log_{10} 2 = 138.7$ s. (2) $C = 0.2\ \mu$F, throws of 12 cm and 6 cm, $t = 2$ s: $R = 14.43\ \text{M}\Omega$.

12.5 Charging and discharging of a capacitor through an inductance

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 319–321 (printed). Syllabus items 15 and 16. Charging equations are tagged (i)(1)–(4) and discharge equations (ii)(1)–(4), following the book's separate numbering. Section 12.6 onwards is not included.

Prerequisites

- Charge and p.d. of a capacitor; self-inductance and back emf (pre-physics P2, P3).
- Simple harmonic motion (pre-math H2).

1. Charging

Let C and L be joined to an emf E through a key (Fig. 12.10), and Q the charge on C at time t.

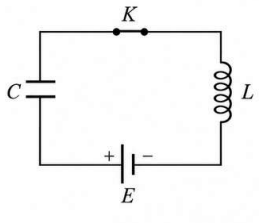


Fig. 12.10 (book): capacitor C and inductance L joined through a Morse key to a cell E.

$$L\frac{dI}{dt} + \frac{Q}{C} = E \qquad (12.5(\text{i})(1))$$

With $I = dQ/dt$, and $Q_0 = CE$, this becomes

$$L\frac{d^2Q}{dt^2} + \frac{Q}{C} = E$$
$$\frac{d^2Q}{dt^2} + \frac{1}{LC}(Q - Q_0) = 0 \qquad (12.5(\text{i})(2))$$

which is simple harmonic motion with period

$$T = 2\pi\sqrt{LC} \qquad (12.5(\text{i})(3))$$

$$\nu = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(\text{i})(4))$$

2. Discharge

With no external emf (Fig. 12.11),

$$\frac{Q}{C} + L\frac{dI}{dt} = 0$$
$$\frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0 \qquad (12.5(\text{ii})(1))$$

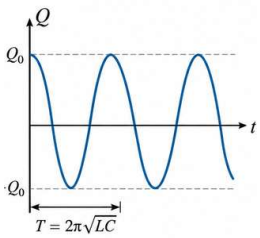
whose general solution, with $\omega = 1/\sqrt{LC}$, is

$$Q = A\sin(\omega t + \theta) \qquad (12.5(\text{ii})(2))$$

With $Q = Q_0$ at $t = 0$ and $\theta = \pi/2$,

$$Q = Q_0 \sin\left(\frac{t}{\sqrt{LC}} + \frac{\pi}{2}\right) \qquad (12.5(\text{ii})(3))$$

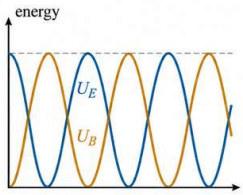
$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(\text{ii})(4))$$



Undamped oscillation of charge, Q = Q_0 sin(t/√LC + π/2), period T = 2π√LC (book, §12.5).

3. Energy

The energy alternates between the electric field of C and the magnetic field of L.



Energy alternates between the electric field of C and the magnetic field of L (book, §12.5).

Section 3: emf of a rotating coil; mean, r.m.s. and virtual values (Murugeshan 13.1)

13.1 EMF induced in a rotating coil

Source: book-sources/m4-s3-ac-emf.html, Murugeshan pp. 329–333 (printed). Syllabus item 17. The book does not number the equations of this section; they are shown unnumbered.

Prerequisites

- Magnetic flux and Faraday's law (pre-physics P4).
- Sinusoidal functions and their integrals (pre-math M5, M7).

1. Set-up

A rectangular coil of N turns, sides a and b, rotates at angular velocity ω about an axis perpendicular to a uniform field $\mathbf{B}$ (Fig. 13.1). The normal makes angle $\theta = \omega t$ with $\mathbf{B}$ at time t, taking $\theta = 0$ at the instant the coil plane is perpendicular to $\mathbf{B}$.

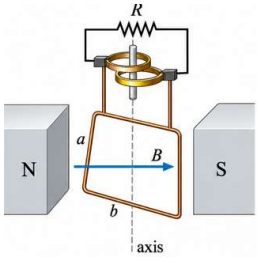


Fig. 13.1 (book): rectangular coil rotating between poles N and S, with slip rings and resistance R.

$$\phi = NBA\cos\omega t, \qquad A = ab$$

2. Induced emf

> **RECALL (PLUS TWO)** Flux $\phi = BA\cos\theta$ and emf $-d\phi/dt$ (Faraday). With N turns and $\theta = \omega t$ the emf is sinusoidal.

$$E = -\frac{d\phi}{dt} = E_0 \sin\omega t, \qquad E_0 = NBA\omega$$

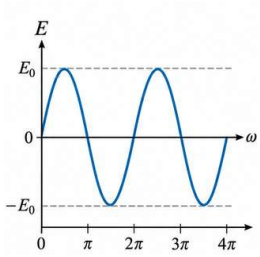


Fig. 13.2 (book): emf E = E_0 sin ωt against ωt; a sine curve.

E_0 is the peak value. The period is $T = 1/\nu$ and $\omega = 2\pi\nu$ (Fig. 13.3).

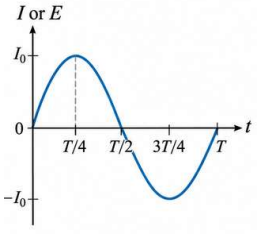
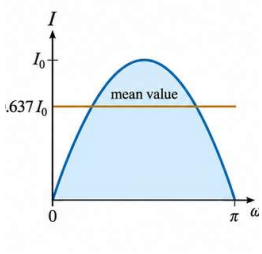


Fig. 13.3 (book): one cycle of ac voltage or current, with peak value and the times T/4, T/2, 3T/4, T.

3. Mean, r.m.s. and virtual values

$$I_{mean} = \frac{2I_0}{\pi} = 0.637\,I_0, \qquad E_{mean} = 0.637\,E_0$$



Mean value over half a cycle, 0.637 I_0 (book, §13.1, shaded area).

$$\overline{I^2} = \frac{I_0^2}{2}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707\,I_0$$

$$E_{rms} = \frac{E_0}{\sqrt{2}}$$

The r.m.s. value equals the virtual (effective) value: the direct current giving the same heat in the same resistance.

4. Form factor and half-cycle mean

$$\text{Form factor} = \frac{I_{rms}}{I_{mean}} = \frac{0.707}{0.637} = 1.11$$

Over half a period, $E_{av} = 2E_0/\pi$.

5. Impedance and reactance

The impedance is $Z = E/I$; reactance is the part due to L or C (X_L, X_C); admittance is $Y = 1/Z$.

Section 4: AC circuits with R, L and C; phasors and the j operator (Murugeshan 13.2)

13.2 Alternating current circuit theory

Source: book-sources/m4-s4-ac-circuits.html, Murugeshan pp. 334–337 (printed). Syllabus items 18, 19 and 20. The parallel circuit, §13.2(vi), is in Section 5 with §13.4. Equations are tagged with the book's numbers.

Prerequisites

- Phase of a sinusoid (pre-physics H2).
- Complex numbers and the phasor (pre-math B1, B2).

1. Pure resistance (i)

> **RECALL (PLUS TWO)** Ohm's law $V = IR$ (Plus Two) gives $I = E/R$ in complex form. The current is in phase with the emf.

For $E = E_0 e^{j\omega t}$ across R (Fig. 13.4), $I = E/R = I_0 e^{j\omega t}$ with $I_0 = E_0/R$. The current is in phase with the voltage (Fig. 13.5).

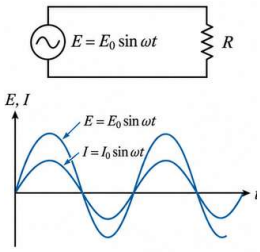


Fig. 13.4 and 13.5 (book): pure resistance; E and I in phase.

2. Pure inductance (ii)

$$I = \frac{E}{j\omega L} = \frac{E_0}{\omega L} e^{j(\omega t - \pi/2)} \qquad (13.2(\text{ii})(1))$$

$$I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right) \qquad (13.2(\text{ii})(2))$$

The current lags the emf by $\pi/2$. The impedance is $j\omega L$.

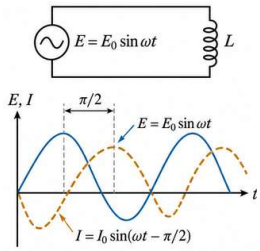


Fig. 13.6 and 13.7 (book): pure inductance; current lags the emf by π/2.

3. Pure capacitance (iii)

From $q/C = E$, $I = dq/dt = j\omega CE$, so $I = I_0 e^{j(\omega t + \pi/2)}$ with $I_0 = E_0/(1/\omega C)$. The current leads the emf by $\pi/2$. The impedance is $1/(j\omega C)$.

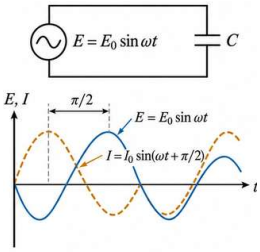


Fig. 13.8 and 13.9 (book): pure capacitance; current leads the emf by π/2.

4. Inductance and resistance in series (iv)

$$I = \frac{E}{R + j\omega L} \qquad (13.2(\text{iv})(1))$$

$$I = \frac{E}{Z} \qquad (13.2(\text{iv})(2))$$

$$Z = R + j\omega L \qquad (13.2(\text{iv})(3))$$

With $\tan\theta = \omega L/R$, $Z = \sqrt{R^2 + \omega^2 L^2}\, e^{j\theta}$, and

$$I = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}} e^{j(\omega t - \theta)} \qquad (13.2(\text{iv})(4))$$

$$I = I_0 e^{j(\omega t - \theta)} \qquad (13.2(\text{iv})(5))$$

$$I_0 = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}} \qquad (13.2(\text{iv})(6))$$

The current lags the emf by θ.

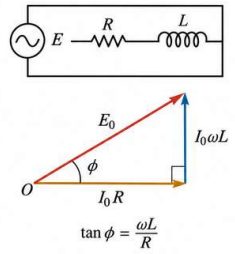


Fig. 13.10 (book): R and L in series, with the phasor diagram of I_0R and $I_0\omega L$.

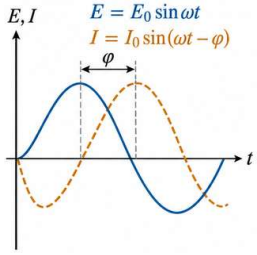


Fig. 13.11 (book): emf and current for R and L in series; current lags by θ.

5. Capacitance and resistance in series (v)

$$I\left(R + \frac{1}{j\omega C}\right) = E = E_0 e^{j\omega t} \quad (13.2(\text{v})(1))$$

$$I = \frac{E}{R + 1/(j\omega C)} \quad (13.2(\text{v})(2))$$

$$I = \frac{E}{Z} \quad (13.2(\text{v})(3))$$

$$I = \frac{E}{R - j/(\omega C)} = \frac{E_0 e^{j\omega t}}{\sqrt{R^2 + 1/(\omega C)^2}}\, e^{-j\theta} \quad (13.2(\text{v})(4))$$

$$\theta = \tan^{-1}\left(\frac{1/\omega C}{R}\right) \quad (13.2(\text{v})(5))$$

$$I = \frac{E_0}{\sqrt{R^2 + (1/\omega C)^2}}\, e^{j(\omega t + \theta)} \quad (13.2(\text{v})(6))$$

The current leads the voltage by θ.

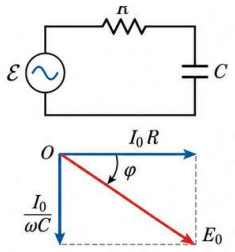


Fig. 13.12 (book): R and C in series, with the phasor diagram.

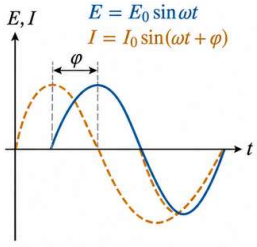


Fig. 13.13 (book): emf and current for R and C in series; current leads by θ.

6. The j operator

j rotates a phasor $90°$ anticlockwise and $-j$ rotates it $90°$ clockwise. The total emf is $E = E_R + jE_L - jE_C$. The complex impedance $Z = R + jX$ has resistance R and effective reactance $X = X_L - X_C$.

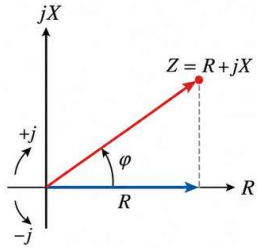


Fig. 13.20 (book): phasors E_R, jE_L and −jE_C; j rotates a phasor by 90°.

13.2 (vi), 13.3 and 13.4 Series and parallel resonance

Source: book-sources/m4-s5-series-parallel.html, Murugeshan pp. 338–347 (printed). Syllabus items 20 and 21. The series derivation is given by two methods: (a) first method, differential equation; (b) second method, complex impedance. Equations are tagged with the book's numbers.

Prerequisites

- Impedance and phase of R, L and C (Section 4).
- Complex numbers and the modulus (pre-math B1, B3).

1. Parallel LC (§13.2(vi))

$$I = I_L + I_C = E_0\left(\omega C - \frac{1}{\omega L}\right)\cos\omega t \qquad (13.2(\text{vi})(1))$$

The current vanishes when $\omega C = 1/\omega L$, giving

$$\nu_0 = \frac{1}{2\pi\sqrt{LC}} \qquad (13.2(\text{vi})(2))$$

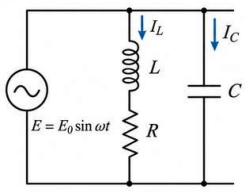


Fig. 13.14 (book): parallel L and C across E = E_0 sin ωt.

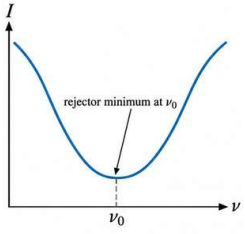


Fig. 13.15 and 13.23 (book): current against frequency, minimum at the parallel resonant frequency.

2. Series LCR, first method (§13.3)

> **RECALL (PLUS TWO)** Series elements add their opposition. Here the reactances subtract, so $Z = \sqrt{R^2 + (X_L - X_C)^2}$.

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{1}{C}\frac{dQ}{dt} = E_0\omega\cos\omega t \qquad (13.3(\text{a})(1))$$

With $I = I_0\sin(\omega t - \phi)$, equating coefficients gives

$$I = I_0\sin(\omega t - \phi) \qquad (13.3(\text{a})(2))$$

$$\left(-L\omega^2 + \frac{1}{C}\right)I_0 = -E_0\omega\sin\phi \qquad (13.3(\text{a})(3))$$

$$R\omega I_0 = E_0\omega\cos\phi \qquad (13.3(\text{a})(4))$$

$$\tan\phi = \frac{\omega L - 1/\omega C}{R} \qquad (13.3(\text{a})(5))$$

$$I_0 = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} \qquad (13.3(\text{a})(6))$$

$$I = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}\sin(\omega t - \phi) \qquad (13.3(\text{a})(7))$$

The current lags if $X_L > X_C$, leads if $X_L < X_C$, and is in phase if $X_L = X_C$.

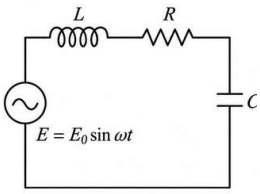
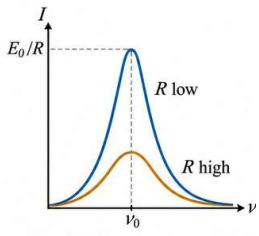


Fig. 13.16 (book): R, L and C in series across E = E_0 sin ωt.

3. Series resonance

The impedance is least, $Z = R$, when $\omega L = 1/\omega C$; the current is then E_0/R. The resonant frequency is $\nu_0 = 1/(2\pi\sqrt{LC})$.

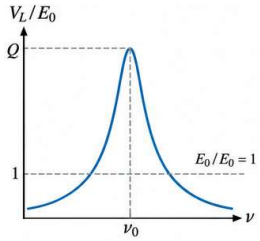


Figs. 13.17 and 13.18 (book): current against frequency at resonance, for low and high R.

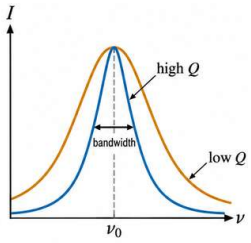
$$I = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{C\omega}\right)^2}} \times \sin(\omega t - \phi) \qquad (13.3(\text{res})(1))$$

4. Q-factor, voltage magnification and bandwidth

At resonance the voltage across L is QE_0, where $Q = \omega_0 L/R$. A larger Q gives a sharper peak.



Voltage magnification at resonance: V_L / E_0 = Q (book, §13.3).



Sharp and flat resonance curves; a large Q gives a narrow bandwidth (book, §13.3).

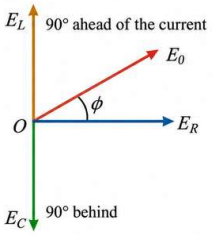


Fig. 13.19 (book): vector diagram of the series LCR circuit.

5. Series LCR, second method (§13.3)

$$E = E_0 e^{j\omega t} \qquad (13.3(b)(1))$$

$$Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2}\, e^{j\phi}$$
$$\tan\phi = \frac{\omega L - 1/\omega C}{R} \qquad (13.3(b)(2))$$

$$I = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}\, e^{j(\omega t - \phi)} \qquad (13.3(b)(3))$$

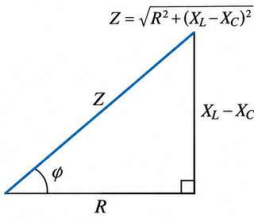
$$I = I_0 e^{j(\omega t - \phi)} \qquad (13.3(b)(4))$$

$$i = \mathrm{Im}(I)$$
$$= \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} \sin(\omega t - \phi) \qquad (13.3(b)(5))$$

The actual current is the imaginary part of the complex current.

6. Parallel resonance: the rejector (§13.4)

C is in parallel with R and L in series. The admittance is least, and the impedance greatest, at $\omega_0 = \sqrt{1/LC - R^2/L^2}$, which is $1/(2\pi\sqrt{LC})$ when R is small. The impedance at resonance is $Z = L/(RC)$.



Impedance triangle for series LCR: Z = √(R² + (X_L − X_C)²) (book, §13.3).

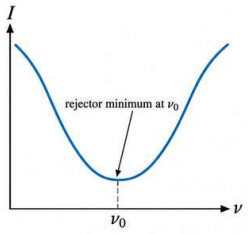


Fig. 13.15 and 13.23 (book): current against frequency, minimum at the parallel resonant frequency.

7. Worked examples

§13.3 Example 1 (printed p. 344): $Z = 53.97\ \Omega$, $I_{rms} = 3.71$ A, $\cos\phi = 0.964$, true power 716 W. §13.3 Example 2 (printed p. 344): resonance at $C = 2\ \mu$F, $I = 1.0$ A, p.d. 100 V across R and 1570 V across L and C. §13.4 Example 1 (printed p. 347): $\nu_0 = 112\,590$ Hz; peak make-up current 15×10^{-6} A.

Section 6: power in AC circuits (Murugeshan 13.5)

13.5 Power in an AC circuit

Source: book-sources/m4-s6-power.html, Murugeshan pp. 347–351 (printed), with Examples 1–5. Syllabus item 22. §13.6 (choke coil) is outside this module; its heading only is shown. Equations (1) and (2) and the Example 2 equations are tagged with the book's numbers.

Prerequisites

- Power in a DC circuit (pre-physics P8).
- Product-to-sum identities and the average of $\cos^2$ over a cycle (pre-math M5, M7).

1. Instantaneous power

> **RECALL (PLUS TWO)** $P = VI = I^2R$ for DC (Plus Two). For AC the average power carries the factor $\cos\phi$.

$$E \times I = \frac{1}{2} E_0 I_0 \left[\cos\phi - \cos(2\omega t - \phi)\right] \qquad (13.5(1))$$

2. Average power

$$P = \frac{1}{2} E_0 I_0 \cos\phi = E_{rms} I_{rms} \cos\phi \quad (13.5(2))$$

The apparent power is $E_{rms} I_{rms}$. The power factor is $\cos\phi$, which equals the true power divided by the apparent power.

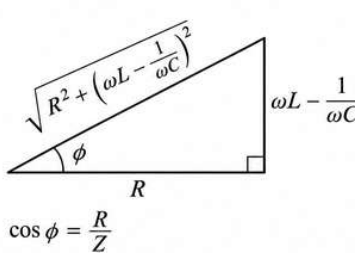


Fig. 13.24 (book): power-factor triangle with sides R, ωL − 1/ωC and Z.

3. Special cases

Purely resistive: $\cos\phi = 1$. Purely inductive or purely capacitive: $\cos\phi = 0$ and the true power is zero.

4. Wattless current

The current is wattless when no power is consumed. A pure inductor or capacitor, with no resistance, draws a wattless current; this is the principle of the choke coil. The component $I\cos\phi$ gives the power and $I\sin\phi$ does not.

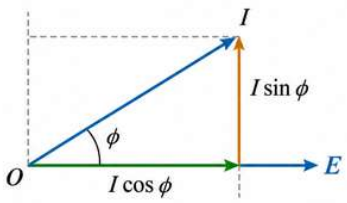
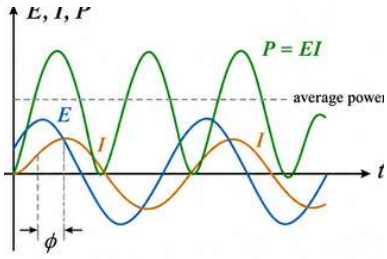


Fig. 13.25 (book): I cos φ (power component) and I sin φ (wattless component).



Emf, current and instantaneous power against time; the average power is shown dashed.

5. Worked examples

Example 1 (printed p. 348): $\cos\phi = 0.062$; power 0.00189 W. Example 2 (printed pp. 348–349): $R = 16.7\ \Omega$, $L = 37.3$ mH, from the equations tagged 13.5(ex2)(1)–(5). Example 3 (printed pp. 350–351): power component 0.20 A, wattless component 0.87 A. Example 4 (printed p. 351): wattmeter reading 514.5 W. Example 5 (printed p. 351): no power is dissipated in a pure inductor.

Example 2 equations

$$\frac{50^2}{[R^2 + (100\pi)^2 L^2]} R = 100\ \text{W} \qquad (13.5(\text{ex2})(1))$$

$$\frac{50^2 R}{[R^2 + (200\pi)^2 L^2]} = 50\ \text{W} \qquad (13.5(\text{ex2})(2$$

$$2500\, R = 100\left[R^2 + 10^4 \pi^2 L^2\right] \qquad (13.5(\text{ex2})(3))$$

$$2500\, R = 50\left[R^2 + 4 \times 10^4 \pi^2 L^2\right] \qquad (13.5(\text{ex2})(4))$$

$$R^2 = 2 \times 10^4 \times \pi^2 L^2$$
$$R = 100\sqrt{2}\,\pi L \qquad (13.5(\text{ex2})(5))$$

§13.6 (choke coil) begins on printed p. 351 and is not part of this module.